





**Parul University**  
**Faculty of Engineering and Technology**  
**Parul Institute of Engineering and Technology**  
**Computer Science & Engineering Department**

| <b>Name of Teacher: Krunal N Patel</b><br><b>Subject Name: Computer Organization and Microprocessor (COMA) (303105210)</b> |                                                              | <b>Hrs./Week: 3</b><br><b>4CSE9</b><br><b>Mon &amp; Thur</b> | <b>Hrs./Week: 3</b><br><b>4CSE10</b><br><b>Tue &amp; Wed</b> |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|
| Sr. No.                                                                                                                    | Name of Topic                                                | Planned Date                                                 | Planned Date                                                 |
| <b>CH 1</b>                                                                                                                | <b>Introduction to Microprocessor 8085</b>                   |                                                              |                                                              |
| 1                                                                                                                          | 1. Microprocessor: Definition                                | 19/01/26                                                     | 20/01/26                                                     |
|                                                                                                                            | 2. Basics of 8085                                            |                                                              |                                                              |
|                                                                                                                            | 3. Difference between microprocessor and microcontroller     |                                                              |                                                              |
|                                                                                                                            | 4. History of microprocessor                                 |                                                              |                                                              |
| 2                                                                                                                          | 1. Instruction set architecture                              | 19/01/26                                                     | 20/01/26                                                     |
|                                                                                                                            | 2. What is machine language?                                 |                                                              |                                                              |
|                                                                                                                            | 3. Explain assembly language.                                |                                                              |                                                              |
|                                                                                                                            | 4. Difference between machine language and assembly language |                                                              |                                                              |
| 3                                                                                                                          | 1. Overview of the 8085 Microprocessor                       | 22/01/26                                                     | 21/01/26                                                     |
|                                                                                                                            | 2. Registers in the 8085 Microprocessor                      |                                                              |                                                              |
|                                                                                                                            | 3. Basics of 8085 Instructions                               |                                                              |                                                              |
| 4                                                                                                                          | 1. Instruction Format                                        | 29/01/26                                                     | 27/01/26                                                     |
|                                                                                                                            | 2. Addressing Modes                                          |                                                              |                                                              |
|                                                                                                                            | 3. Memory Hierarchy and Storage                              |                                                              |                                                              |
|                                                                                                                            | 4. Instruction Storage and Fetching                          |                                                              |                                                              |
| <b>CH2</b>                                                                                                                 | <b>Microprocessor architecture and interfacing</b>           |                                                              |                                                              |
| 5                                                                                                                          | 1. Definition and Basic Concepts                             | 2/02/26                                                      | 27/01/26                                                     |

|            |                                                                               |          |          |
|------------|-------------------------------------------------------------------------------|----------|----------|
|            | 2. Key Components of a Microprocessor                                         |          |          |
|            | 3. Core Components: ALU (Arithmetic Logic Unit), CU (Control Unit), Registers |          |          |
|            | 4. Memory Architecture                                                        |          |          |
|            | 5. Bus Architecture                                                           |          |          |
| 6          | 1. Microprocessor operations                                                  | 2/02/26  | 28/01/26 |
| 7          | 1. Memory devices                                                             | 5/02/26  | 3/02/26  |
|            | 2. Types of Memory Devices                                                    |          |          |
|            | 3. Memory Architecture                                                        |          |          |
| 8          | 1. I/O devices                                                                | 9/02/26  | 3/02/26  |
|            | 2. Types of I/O devices                                                       |          |          |
|            | 3. I/O devices Architecture                                                   |          |          |
| 9          | 1. Memory interfacing                                                         | 9/02/26  | 4/02/26  |
|            | 2. Memory size calculation                                                    |          |          |
| 10         | 1. Find the address of the memory                                             | 12/02/26 | 10/02/26 |
|            | 2. Examples of memory interfacing                                             |          |          |
| 11         | 1. Interfacing I/O devices                                                    | 16/02/26 | 10/02/26 |
| 12         | 1. Examples of I/O devices Interfacing                                        | 16/02/26 | 11/02/26 |
| <b>CH3</b> | <b>Programming methods with Instructions</b>                                  |          |          |
| 13         | 1. 8085 Instructions                                                          | 19/02/26 | 18/02/26 |
|            | 2. Data Transfer Instructions                                                 |          |          |
| 14         | 1. Arithmetic Instructions: Basics                                            | 23/02/26 | 24/02/26 |
|            | 2. List of Arithmetic Instructions                                            |          |          |
|            | 3. Addition Instructions with Examples                                        |          |          |

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|     | 4. Substraction Instructions with Examples        |          |          |
|     | 5. Increment/Decrement Instructions with Examples |          |          |
| 15  | 1. Concept of Loop                                | 23/02/26 | 24/02/26 |
|     | 2. Classification of Loop                         |          |          |
|     | 3. Design of Loop                                 |          |          |
|     | 4. Unconditinal Loop                              |          |          |
|     | 5. Conditinal Loop                                |          |          |
|     | 6. Nested Loop                                    |          |          |
| 16  | 1. Introduction of indexing                       | 26/02/26 | 25/02/26 |
|     | 2. Looping flow chart                             |          |          |
|     | 3. Concept of counting                            |          |          |
|     | 4. Introduction to logical instructions           |          |          |
|     | 5. List and explanation of logical instructions   |          |          |
| 17  | 1. Compare instructions                           | 2/03/26  | 3/03/26  |
|     | 2. Rotate instructions                            |          |          |
| CH4 | Additional Programming techniques                 |          |          |
| 18  | 1. Introduction to counter and flowchart          | 2/03/26  | 3/03/26  |
|     | 2. Counter Design with time delay                 |          |          |
|     | 3. Introduction to time delay and flowchart       |          |          |
|     | 4. Calculation of time delay                      |          |          |
|     | 5. Time Delay using one register                  |          |          |
| 19  | 1. Time Delay using register pair                 | 5/03/26  | 10/03/26 |
|     | 2. Time Delay using nested Loop                   |          |          |
| 20  | 1. Concepts of Stack                              | 9/03/26  | 10/03/26 |

|     |                                                    |          |          |
|-----|----------------------------------------------------|----------|----------|
|     | 2. Operation of Stack                              |          |          |
|     | 3. Instructions used for stack                     |          |          |
|     | 4. Programing based on stack concepts              |          |          |
| 21  | 1. Introduction to Subroutines                     | 9/03/26  | 11/03/26 |
|     | 2. Brief of Subrotines concept                     |          |          |
|     | 3. Programing techniques using subroutine          |          |          |
| 22  | 1. Introduction to Restart Instructions            | 12/03/26 | 17/03/26 |
|     | 2. Operation and Functionality                     |          |          |
|     | 3. Execution Process & Uses                        |          |          |
| 23  | 1. CALL Instruction                                | 16/03/26 | 17/03/26 |
|     | 2. RETURN Instruction                              |          |          |
|     | 3. Stack Operations in CALL/RET                    |          |          |
| 24  | 1. Binary to BCD code conversion                   | 16/03/26 | 18/03/26 |
|     | 2. BCD to Binary code conversion                   |          |          |
| 25  | 1. ASCII to Binary Code conversion                 | 19/03/26 | 31/03/26 |
|     | 2. Binary to ASCII code conversion                 |          |          |
| CH5 | 8085 Interrupts                                    |          |          |
| 26  | 1. Introduction to Interrupts                      | 30/03/26 | 31/03/26 |
|     | 2. Types of Interrupts in 8085                     |          |          |
|     | 3. Interrupt Pins and Priority Structure           |          |          |
|     | 4. Enabling, Disabling, and Handling Interrupts    |          |          |
| 27  | 1. Concept of Vectored and Non-Vectored Interrupts | 30/03/26 | 1/04/26  |
|     | 2. Vectored Interrupt Handling in 8085             |          |          |
|     | 3. Non-Vectored Interrupt Handling in 8085         |          |          |

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|-----|-----------------------------------------------------|----------|----------|
| 28  | 1. Definition of Latency Time                       | 2/04/26  | 7/04/26  |
|     | 2. Definition of Response Time                      |          |          |
|     | 3. Comparison and Significance                      |          |          |
| 29  | 1. Introduction to Multiple Interrupts              | 6/04/26  | 7/04/26  |
|     | 2. Interrupt Priority and Nesting                   |          |          |
|     | 3. Hardware and Software Techniques                 |          |          |
|     | 4. Designing Interrupt Handling Routines            |          |          |
| CH6 | Computer Organization                               |          |          |
| 30  | 1. Basics of Register Transfer Language (RTL)       | 6/04/26  | 8/04/26  |
|     | 2. RTL Notation and Operations                      |          |          |
|     | 3. Bus Design Using Multiplexers                    |          |          |
|     | 4. Bus Design Using Tri-State Buffers               |          |          |
| 31  | 1. Memory Transfer Basics                           | 9/04/26  | 15/04/26 |
|     | 2. Control Signals for Memory Transfers             |          |          |
|     | 3. Types of Arithmetic Micro-Operations             |          |          |
|     | 4. RTL Representation of Arithmetic Operations      |          |          |
| 32  | 1. Introduction to Logic Micro-Operations in 8085   | 13/04/26 | 21/04/26 |
|     | 2. Implementation of Logic Micro-Operations in 8085 |          |          |
|     | 3. Introduction to Shift Micro-Operations in 8085   |          |          |
|     | 4. Application of Shift Micro-Operations in 8085    |          |          |
| 33  | 1. Arithmetic logical shift unit                    | 13/04/26 | 21/04/26 |
| 34  | 1. Introduction to Instruction Codes                | 16/04/26 | 22/04/26 |
|     | 2. Types of Computer Registers                      |          |          |
|     | 3. Computer Instructions and Their Formats          |          |          |
|     | 4. Execution of Computer Instructions               |          |          |

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|     |                                                                      |          |          |
| 35  | 1. Introduction to Timing and Control in a CPU                       | 20/04/26 | 28/04/26 |
|     | 2. Instruction Cycle and Its Phases                                  |          |          |
|     | 3. Memory Reference Instructions                                     |          |          |
|     | 4. Application of Timing, Control, and Memory Reference Instructions |          |          |
| 36  | 1. Introduction to Register Reference Instructions                   | 20/04/26 | 28/04/26 |
|     | 2. Execution of Register Reference Instructions                      |          |          |
|     | 3. Introduction to I/O Reference Instructions                        |          |          |
|     | 4. Application of I/O and Register Reference Instructions            |          |          |
| 37  | 1. Introduction to Interrupts in Microprocessors                     | 23/04/26 | 29/04/26 |
|     | 2. Interrupt Handling and Control Flow                               |          |          |
|     | 3. Introduction to the Design of an Accumulator Unit                 |          |          |
|     | 4. Design and Implementation of the Accumulator Unit                 |          |          |
| CH7 | Computer Organization                                                |          |          |
| 38  | 1. Introduction to Machine Language                                  | 27/04/26 | 29/04/26 |
|     | 2. Basics of Assembly Language Programming                           |          |          |
|     | 3. Comparison and Translation between Machine and Assembly Language  |          |          |
|     | 4. Writing and Executing Simple Assembly Programs                    |          |          |
| 39  | 1. Assembler, Program loops                                          | 27/04/26 | 5/05/26  |
| 40  | 1. Programming Arithmetic and logic operations                       | 30/04/26 | 5/05/26  |
| 41  | 1. Introduction to Subroutines in Assembly Language                  | 04/05/26 | 5/05/26  |
|     | 2. Execution and Nesting of Subroutines                              |          |          |
|     | 3. Introduction to I/O Programming Techniques                        |          |          |

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|    | 4. Writing Programs Using Subroutines and I/O Operations           |          |         |
| 42 | 1. Introduction to Memory Hierarchy                                | 04/05/26 | 6/05/26 |
|    | 2. Characteristics and Organization of Memory Levels               |          |         |
|    | 3. Structure and Operation of Main Memory (RAM)                    |          |         |
|    | 4. Analyze Main Memory's Role in the Memory Hierarchy              |          |         |
| 43 | 1. Introduction to Auxiliary Memory                                | 07/05/26 | 6/05/26 |
|    | 2. Characteristics and Performance of Auxiliary Memory             |          |         |
|    | 3. Introduction to Flash Memory Technology                         |          |         |
|    | 4. Analyze Flash Memory vs Other Storage Devices                   |          |         |
| 44 | 1. Introduction to Associative Memory (Content Addressable Memory) | 07/05/26 | 6/05/26 |
|    | 2. Organization and Operation of Associative Memory                |          |         |
|    | 3. Introduction to Cache Memory and Its Need                       |          |         |
|    | 4. Cache Organization                                              |          |         |
| 45 | 1. Virtual memory                                                  | 07/05/26 | 6/05/26 |